

Robust adaptive control of UAV

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Abstract: Aiming at the problems of UAV dynamic modeling error and flight interference, robust control algorithm is introduced in this paper. Based on the establishment of UAV motion equation, the state equation of robust control is obtained by linearization, then the weight function is selected and the controller is designed. MATLAB simulation results show that Compared with PI control, robust longitudinal control of UAV has the characteristics of fast response, small overshoot, short adjustment time and insensitive to modeling error, which shows good control performance.

Key Words: Robust control, linearization, weight function, MATLAB simulation

1 Introduction

With the development of science and technology, unmanned aerial vehicle (UAV), a high-tech product with great use value in military and civil fields, is increasingly attracting the attention of all countries in the world. Compared with manned aircraft, UAV has the characteristics of light weight, small size, flexible mobility, low cost, strong battlefield survival ability, and has high cost performance [1]. It is mainly used in air reconnaissance, ground, attack, topographic, mapping, communication relay and so on [2]. The motion of UAV has the characteristics of strong nonlinearity, strong coupling and high maneuverability [2]. Therefore, UAV has high requirements for control. The stability, accuracy and rapidity of control response are important indexes to measure the control performance of UAV.

There are commonly used methods of UAV control: PID control, synovial control, fuzzy control, adaptive control. PID control algorithm is simple and parameters are fixed. However, it has poor resistance to random disturbances and is not suitable for UAV motion control with strong nonlinear characteristics. The response speed of synovial control is fast, which can effectively overcome the uncertainty of the system. However, due to the approaching speed inertia of the system, the controller has jitter. The fuzzy control has good control effect on the strong nonlinear system and the system which is difficult to express by mathematical model. However, the selection of control rules and membership function needs professionals

to determine; Adaptive control adjusts the controller through the data of system operation. However, it has high requirements for the anti-interference of the control object. The robust control meets the design requirements of the uncertainty of dynamic modeling and external disturbance. Considering the characteristics of UAV and the requirements of control performance. In this paper, robust control [47] is introduced into UAV longitudinal attitude control, to achieve accurate and rapid control of UAV.

2 UAV Modeling

Suppose the UAV is a rigid body, the center of gravity of UAV is set as the origin of aircraft system, the earth's axis system is regarded as an inertial coordinate system, then the 6-DOF longitudinal flight dynamic motion equation of UAV can be described as:

$$\begin{cases} F_x = T \cos(\alpha + \varphi_T) - D - G \sin \gamma \\ F_z = T \sin(\alpha + \varphi_T) + L - G \cos \gamma \\ I_y \frac{dq}{dt} = M^a + T z_T \end{cases} \quad (1)$$

Where M^a is the pitching moment except the engine. $T z_T$ is the pitch torque generated by the engine. γ is track inclination. α is the angle of attack. φ_T is the angle between engine thrust and fuselage longitudinal axis. L , D and T represent lift, drag and thrust respectively. Then, the state equation of UAV longitudinal motion is obtained by linearizing (1) by small disturbance principle, i.e.

$$\begin{cases} \dot{X} = AX + BU \\ Y = CX + D \end{cases} \quad (2)$$

Where A is the coefficient matrix of the internal state relations of the system, i.e.

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$$A = \begin{bmatrix} X_V & X_\alpha & 0 & X_\theta \\ Z_V & Z_\alpha & Z_q & Z_\theta \\ M_V & M_\alpha & M_q & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \quad (3)$$

B represents the input matrix of input to state, i.e.

$$B = \begin{bmatrix} X_{\delta_p} & 0 \\ Z_{\delta_p} & Z_{\delta_e} \\ M_{\delta_p} & M_{\delta_e} \\ 0 & 0 \end{bmatrix} \quad (4)$$

State vector $X = [\Delta V \ \Delta \alpha \ \Delta q \ \Delta \theta]^T$, input vector $U = [\Delta \delta_p \ \Delta \delta_e]^T$, the derivation and meaning of other formulas are shown in reference [8].

3 Controller Design

3.1 Robust Control Method

The general structure of robust control is shown in Fig. 1, Where $G(s)$ is the augmented object model [9], $K(s)$ represents the controller model.

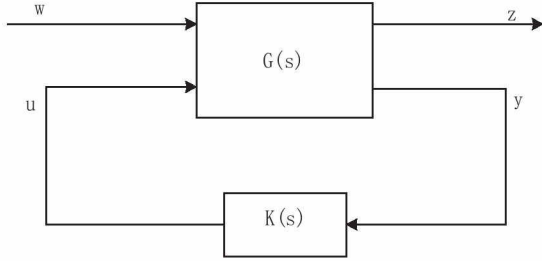


Fig 1. Basic structure of robust control

The state space of the extended object model $G(s)$ is generally expressed as formula (5).

$$G(s) = \begin{bmatrix} G_{11}(s) & G_{12}(s) \\ G_{21}(s) & G_{22}(s) \end{bmatrix} = \begin{bmatrix} A & B_1 & B_2 \\ C_1 & D_{11} & D_{12} \\ C_1 & D_{21} & D_{22} \end{bmatrix} \quad (5)$$

The closed-loop transfer function from w to z is formula (6).

$$T_{zw}(s) = G_{11} + G_{12}K(I - G_{22}K)^{-1}G_{21} \quad (6)$$

The weighted sensitivity function block diagram is shown in the figure

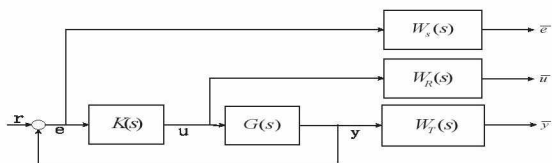


Fig 2. Weighted robust control structure

At this time, the weighted augmented plant P is

$$P = \begin{bmatrix} W_s & -W_s G \\ 0 & W_R \\ 0 & W_T G \\ I & -G \end{bmatrix} = \begin{bmatrix} A & B_1 & B_2 \\ C_1 & D_{11} & D_{12} \\ C_2 & D_{21} & D_{22} \end{bmatrix} \quad (7)$$

Let the number modulus of the forward loop of the weighted function be $L(s) = G(s)K(s)$. Then the corresponding sensitivity function $S(s)$, transfer function $R(s)$, The complementary sensitivity functions $F(s)$ are as follows

$$\begin{cases} S(s) = [I + L(s)]^{-1} \\ R(s) = K(s)[I + L(s)]^{-1} \\ T(s) = G(s)K(s)[I + L(s)]^{-1} \end{cases} \quad (8)$$

The corresponding transfer function matrix from input w to output Z becomes

$$T_{zw} = \begin{bmatrix} W_s S \\ W_R R \\ W_T T \end{bmatrix} \quad (9)$$

3.2 The Choice of Weighting Function

In terms of robust control, the selection of weighting function largely determines the performance of the controller. It is necessary to select the appropriate value of the weighting function through constant adjustment and trial. There are some selection principles as follows: $W_s(s)$ should have integral characteristic and be able to restrain certain interference. Therefore, a larger amplitude should be selected, generally 0.2 ~ 0.8 is appropriate. $W_R(s)$ can limit the size of the control quantity to prevent over saturation and damage to the controller. Therefore, a larger gain can be selected. However, if the gain is too large, the shear frequency will decrease, so the gain is expected to be smaller. It reflects the requirement of robust stability. It should have the ability of restraining modeling error in high frequency band. The weighted sensitivity function should also satisfy the following conditions [10].

$$\begin{cases} \bar{\sigma}(S(s)) \leq \bar{\sigma}[W_s^{-1}(s)] \\ \bar{\sigma}(R(s)) \leq \bar{\sigma}[W_R^{-1}(s)] \\ \bar{\sigma}(T(s)) \leq \bar{\sigma}[W_T^{-1}(s)] \end{cases} \quad (10)$$

To ensure that $W_s(s)$ and $W_T(s)$ bands do not coincide, it should meet the following conditions considering the above

$$\bar{\sigma}(W_s^{-1}(s)) + \bar{\sigma}(W_T^{-1}(s)) \geq 1 \quad (11)$$

Considering the above conditions, the controller weight function designed in this paper is

$$\begin{cases} W_s(s) = \frac{0.0121s + 0.4}{12.89s + 0.024} \\ W_R(s) = 1e - 6 \\ W_T(s) = \frac{s}{10000} \end{cases} \quad (12)$$

Next, whether the selection of weight function meets the requirements is verified. The selection of sensitivity function and complementary sensitivity function is related to the design of the controller. Therefore, on the basis of designing the controller in the end of this paper, the value of weight function is adjusted in turn to achieve better control effect.

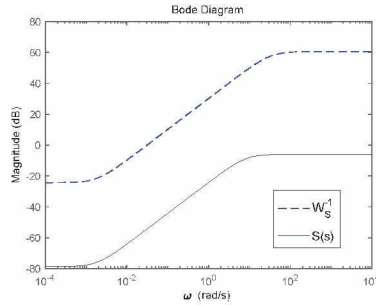


Fig 3. Sensitivity function and Amplitude frequency characteristic curve

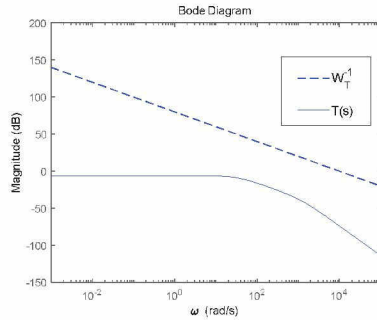


Fig 4. Complementary sensitivity function and amplitude frequency characteristic curve

It can be seen from Figure 3 that the maximum singular values of sensitivity function $S(s)$ are in W_s^{-1} in the whole frequency domain, and it has low pass property. The maximum singular value of the complementary sensitivity function $T(s)$ is also in W_T^{-1} , and it has high pass property. The selection condition of weight function is satisfied.

3.3 Design of Robust Controller

After the weight function is selected, the problem of robust controller design can be reduced to designing a controller $K(s)$ such that $\|T_{zw}(s)\|_\infty < \gamma$ after introducing the weight function, the equation of state is

$$\begin{cases} \dot{x}(t) = A_f x(t) - ZLu(t) \\ y(t) = Kx(t) \end{cases} \quad (13)$$

The relevant coefficients are given in [11]. The unknown parameters and can be solved by the following Riccati equation:

$$\begin{cases} A^T X + XA + X(\gamma^{-2} B_1 B_1^T - B_2 B_2^T) X + C_1^T C_1 = 0 \\ AY + YA^T + Y(\gamma^{-2} C_1^T C_1 - C_2^T C_2) Y + B_1^T B_1 = 0 \end{cases} \quad (14)$$

Then under the premise of the existence of robust controller, the minimum value is found by MATLAB to solve the optimal robust controller.

4 Matlab Simulation Result

After the weight function is designed, the data in reference [12] is used to simulate the aircraft longitudinal parameters.

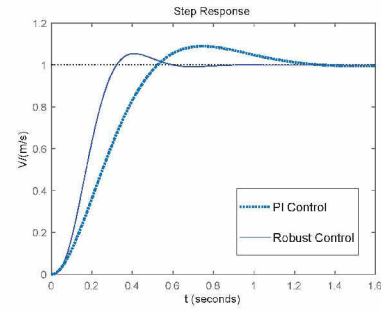


Fig 5. Speed step response

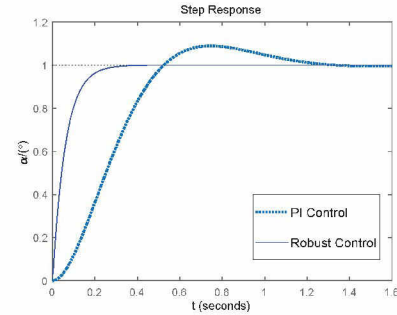


Fig 6. Angle of attack step responses

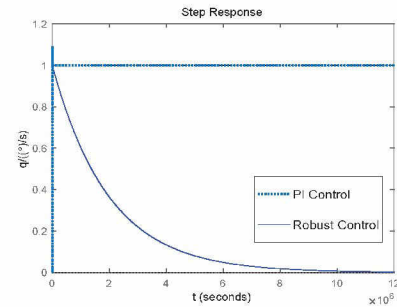


Fig 7. Step response of pitch angular rate

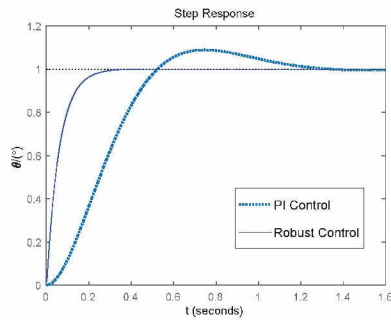


Fig 8. Pitch angle step response

The simulation results show that the robust control can track the input signal better than the PI control when the system input is a step signal. In the speed step response of Figure 5, compared with PI control, robust control has good steady-state performance such as quick response, fast rise time, small overshoot, short over process time; In the step response of angle of attack in Figure 6 and the step response of pitch angle in Figure 8, Angle of attack and pitch angle have stronger tracking performance on the input signal, short rise time and no overshoot, which shows good control performance. In the step response of pitch angular rate in Figure 7 the pitch angle gradually tends to be stable with the change of input signal, and the corresponding pitch angle change rate also gradually tends to 0, which meets the design requirements.

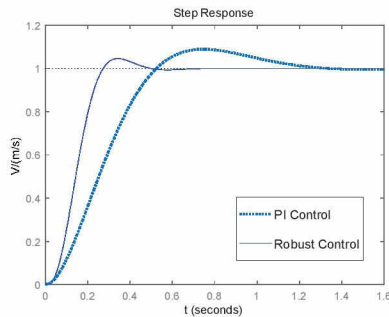


Fig 9. Velocity step response with changing digital and analog parameters

The velocity step response in Figure 9 simulation diagram obtained by changing some parameters in the dynamic model. Compared with PI control, the robust control in the graph has the advantages of fast response, small overshoot, fast rise time and small oscillation times. Compared with Figure 5, there is not much change, showing good robustness. It also shows that robust control

still has good control performance when there are errors in dynamic modeling.

5 Conclusion

This article starts from the perspective of the dynamic modeling error of aircraft longitudinal control and its influence on control, introduced robust hybrid sensitivity control with good robustness. After the aircraft longitudinal dynamic equation is established, the state equation is obtained by linearizing it. Then after the selection of the weight function and the controller design, the simulation is carried out through MATLAB. The simulation results show that the robust hybrid sensitivity control has better steady-state performance than PI control. After modifying some state equation parameters, the simulation results show that robust hybrid sensitivity control can suppress modeling error and has good control performance.

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